

Geospatial Analysis (partial notes)

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How a coastline 100 million years ago
influences modern election results in Alabama

Cretaceous Sediments



Fertile Blackland
Prairie Soil



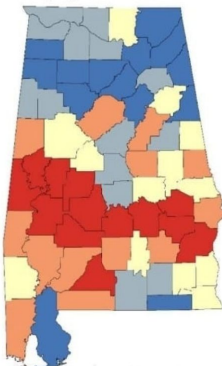
Average Farm Size, 1997



Slave Population, 1860



Black population, 2010



-Starkey Comics

Election Results, 2020



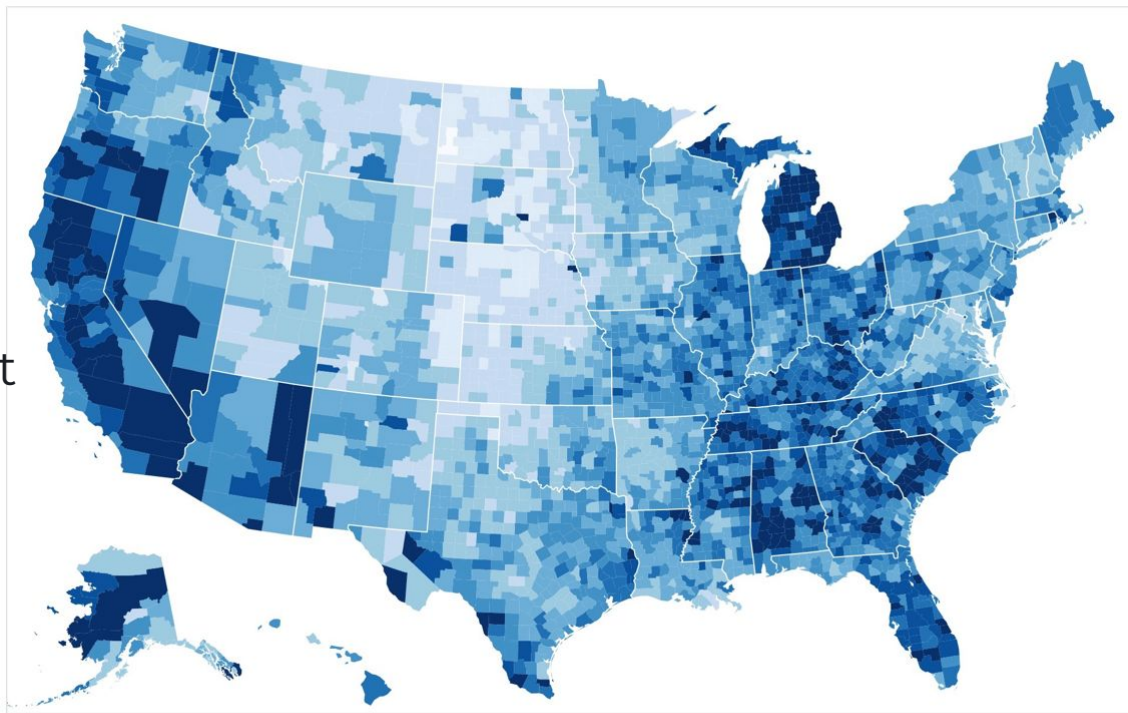
Why Geospatial Analysis?

“Everything is related to everything else, but near things are more related than distant things.” -Tobler 1979

“...the purpose of geographic inquiry is to examine relationships between geographic features collectively and to use the relationships to describe the real-world phenomena that map features represent” -Clarke 2001

Visualizing Geospatial Data

Unemployment
rate by county
(August 2016)



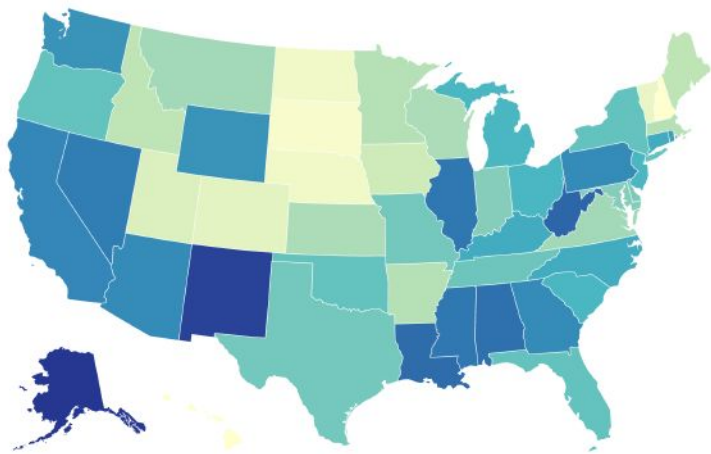
This choropleth encodes unemployment rates from 2008 with a [quantize scale](#) ranging from 0 to 15%. A [threshold scale](#) is a useful alternative for coloring arbitrary ranges.

[Open in a new window.](#)

Choropleth maps are useful for visualizing *clear regional patterns* in the data

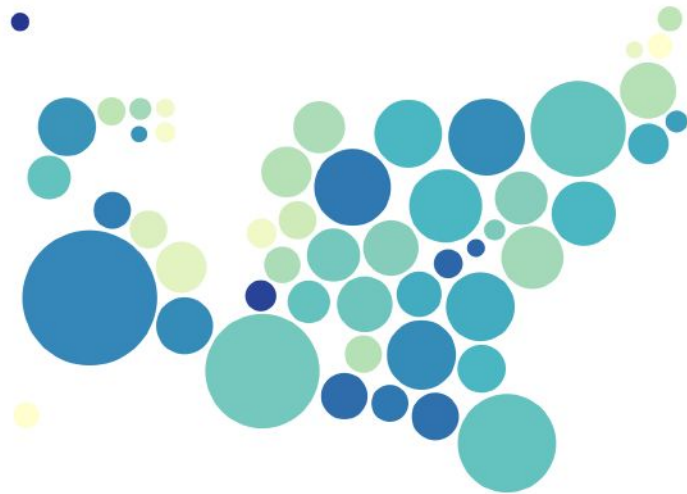
Cartograms should be considered when displaying how many people were affected

NOT IDEAL



Choropleths answer "How much area was affected?"

BETTER



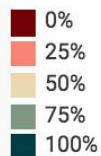
Cartograms answer "How many people were affected?"

Share of individuals using the internet, 2015

Share of individuals using the internet, measured as the percentage of the population. Internet users are individuals who have used the Internet (from any location) in the last 3 months. The Internet can be used via a computer, mobile phone, personal digital assistant, games machine, digital TV etc.

- ☐ **A** Thought about it and have a thought
- ☐ **B** Thought about it and have no thought
- ☐ **C** I'm confused

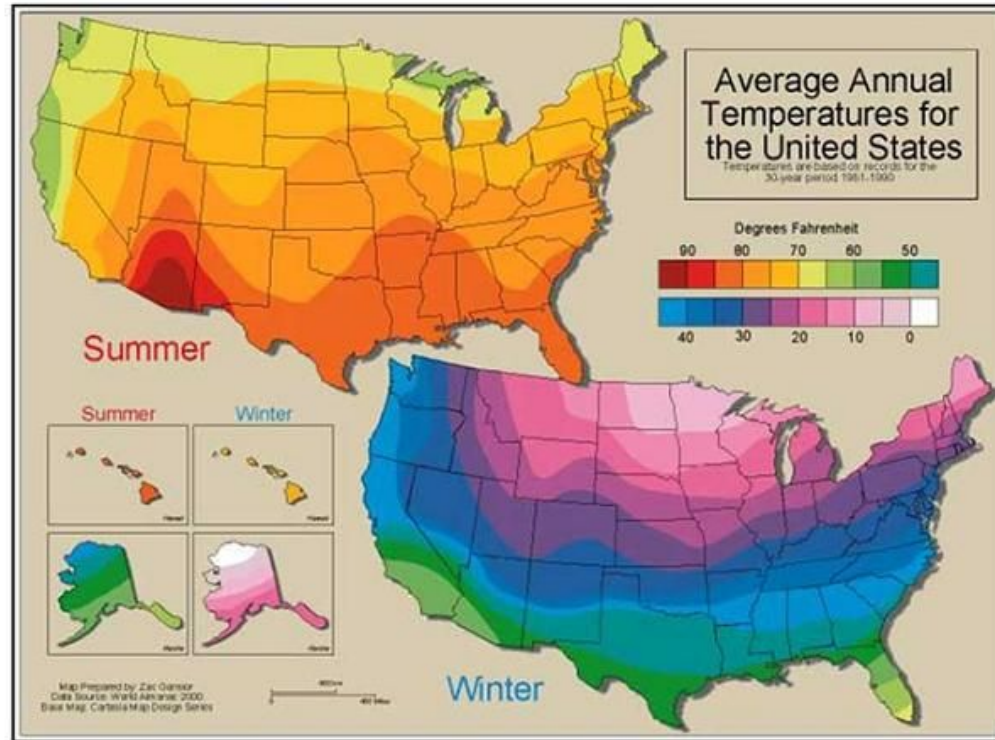
Share of people using the internet



Source: [Our World in Data](#) • [Get the data](#)

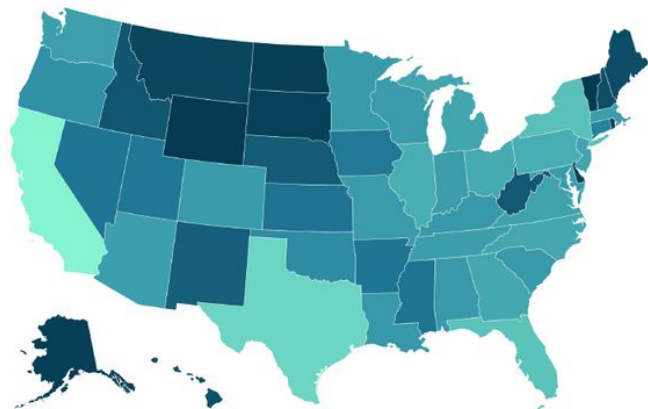


Isarithmic maps demonstrate smooth, continuous phenomena (temperature, elevation, rainfall, etc.)



Use light colors for low values. Dark colors for high values.

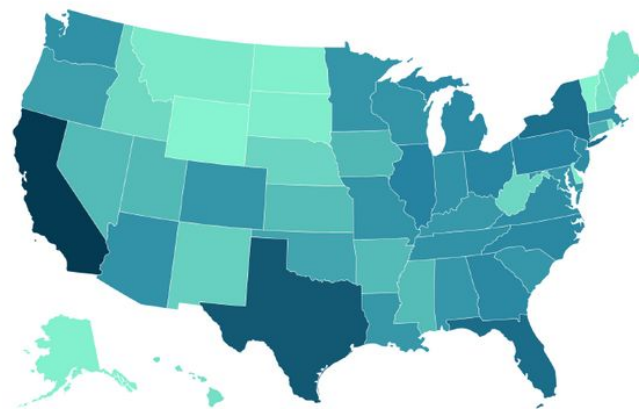
NOT IDEAL



LOW POPULATION HIGH



BETTER



LOW POPULATION HIGH



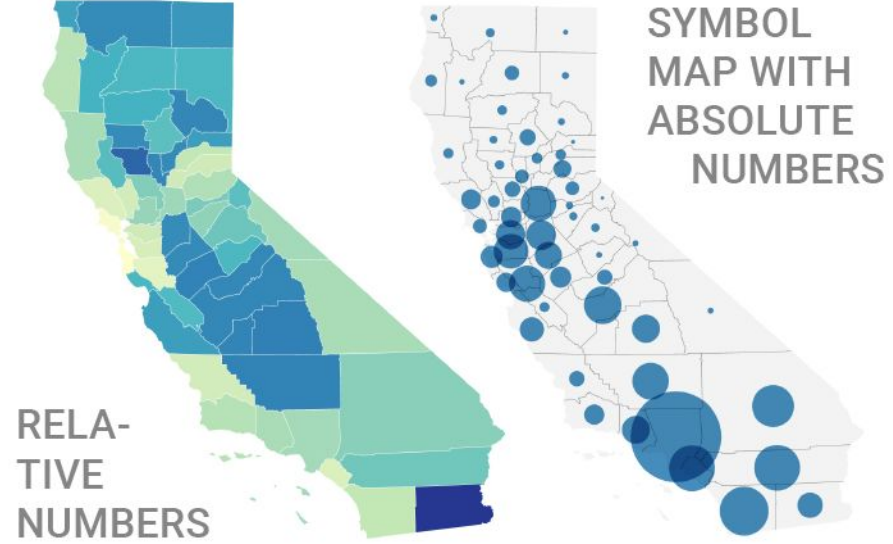
Choropleth maps are useful for visualizing *clear regional patterns* in the data

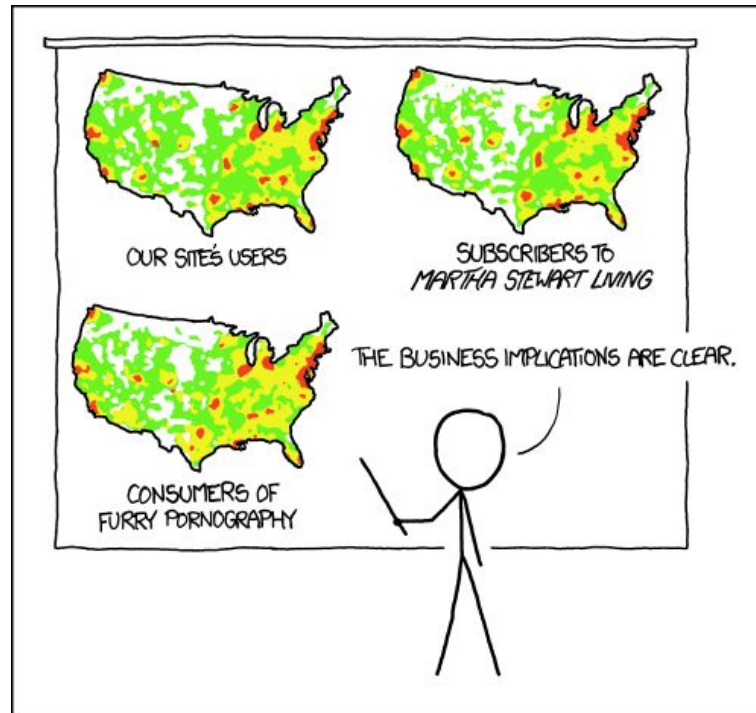
Choropleth should display relative differences, *not* absolute numbers

NOT IDEAL



BETTER



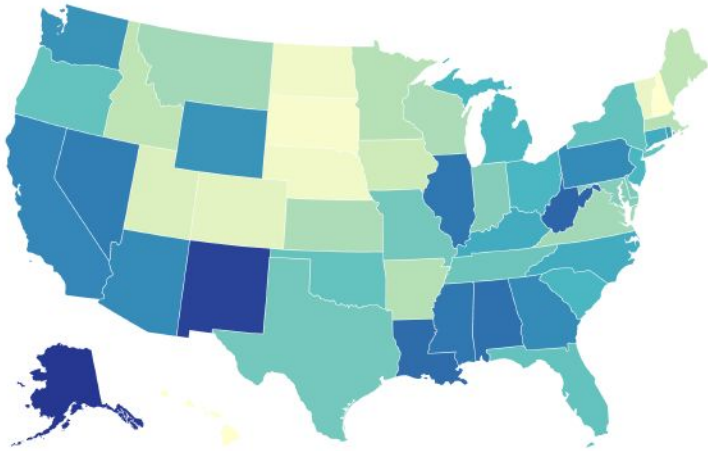


PET PEEVE #208:
GEOGRAPHIC PROFILE MAPS WHICH ARE
BASICALLY JUST POPULATION MAPS

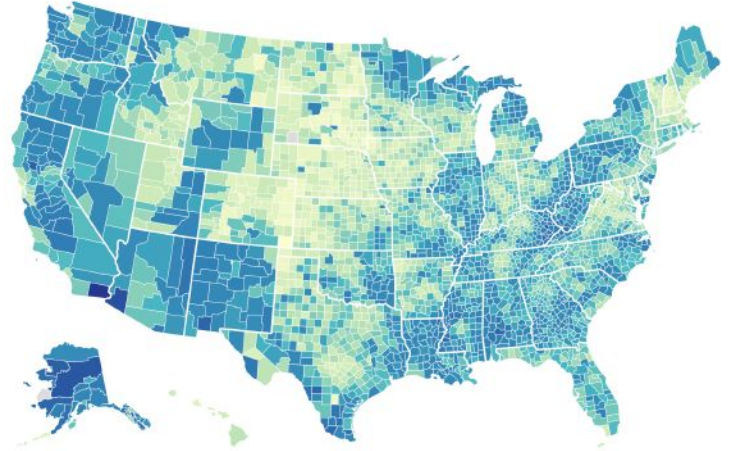
Choropleth maps can be misleading

Consider using the smallest unit possible (but there are exceptions!)

NOT IDEAL



BETTER

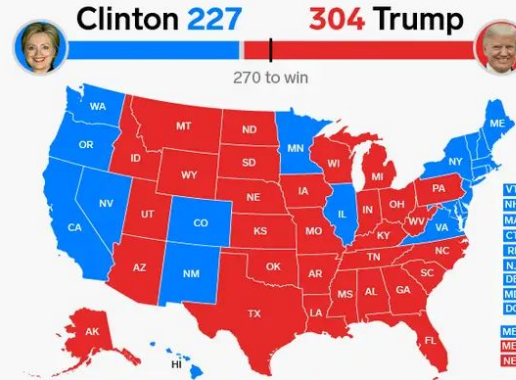


Presidential results 2020 vs. 2016

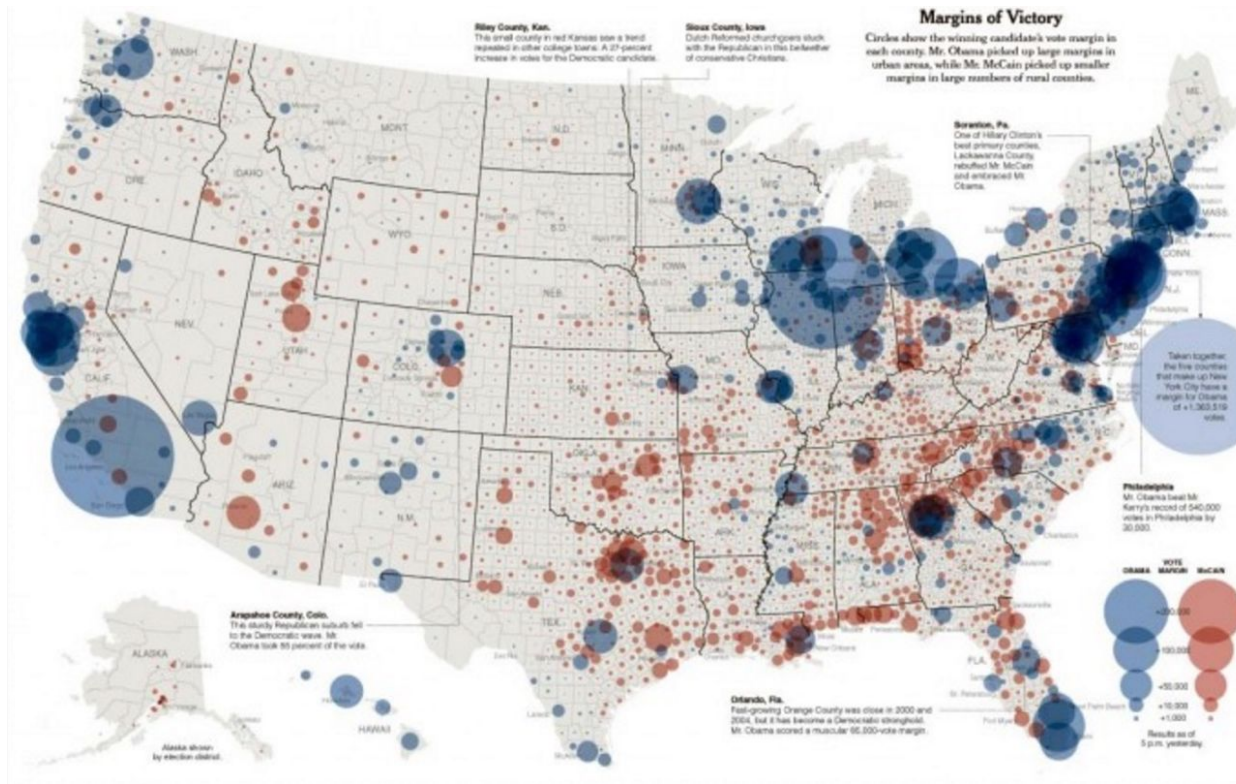
2020



2016



Sometimes summarizing at the state level is ok...



This **bubble graph** more accurately tells the full story, since the size of the bubbles is reflective of the population

...but similar data *can* be displayed more effectively and informatively.

Map: Where Are Students Attending Charter Schools?



The majority of California's charter school student population is concentrated in Los Angeles, San Diego and Bay Area counties. Hover through the counties on each map for more information on their

